Communication Systems: Lectures 13, 17 & 18 Summary

Lecture 13: Bandpass Signals

Objective

Represent a complex bandpass signal centered around carrier frequency f_c as a simple baseband signal from 0 to B Hz. This simplifies signal processing by working with the complex envelope $g(t)$ instead of processing at high frequencies.

Complex Envelope Representation

Complex Form:
$g(t) = x(t) + j \cdot y(t)$

Polar Form:
$g(t) = R(t) \cdot e^{j\theta(t)}$

Where:
- $R(t) = |g(t)| = \sqrt{x^2(t) + y^2(t)}$ - Envelope (amplitude)
- $\theta(t) = \arg[g(t)] = \tan^{-1}\left(\frac{y(t)}{x(t)}\right)$ - Phase

In-Phase and Quadrature Components:
- $x(t) = \text{Re}[g(t)]$ - In-phase component
- $y(t) = \text{Im}[g(t)]$ - Quadrature component

Bandpass Signal Representation

The bandpass signal can be expressed as:
$s(t) = \text{Re}[g(t) \cdot e^{j2\pi f_c t}]$

Expanding:
$s(t) = x(t)\cos(2\pi f_c t) - y(t)\sin(2\pi f_c t)$

Or in polar form:
$s(t) = R(t)\cos(2\pi f_c t + \theta(t))$

Spectrum Relationship

The spectrum of the bandpass signal $S(f)$ is related to the complex envelope spectrum $G(f)$:

$S(f) = \frac{1}{2}[G(f - f_c) + G^*(f + f_c)]$

Where $G^*(f)$ is the complex conjugate of $G(f)$.

Power Calculations

Average Power:
$P_{avg} = \frac{1}{2}\overline{|g(t)|^2} = \frac{1}{2}\overline{R^2(t)}$

Peak Envelope Power (PEP):
$PEP = \frac{1}{2}[\max R(t)]^2$

Power in terms of components:
$P_{avg} = \frac{1}{2}[\overline{x^2(t)} + \overline{y^2(t)}]$

Lecture 17: Amplitude Modulation (AM)

Complex Envelope for AM

For AM signals, the complex envelope is:
$g(t) = A_c[1 + m(t)]$

Where:
- A_c = Carrier amplitude
- $m(t)$ = Message signal (modulating signal)

Key Property: For AM, $Q(t) = 0$ (quadrature component is zero), meaning information cannot be restored using phase, only amplitude.

AM Signal Representation

Time Domain:
$s_{AM}(t) = A_c[1 + m(t)]\cos(2\pi f_c t)$

Envelope:
$R(t) = A_c[1 + m(t)]$

Modulation Parameters

Maximum and Minimum Amplitude:
- $A_{max} = A_c[1 + \max m(t)]$
- $A_{min} = A_c[1 + \min m(t)]$

Modulation Index (Modulation Percentage):
$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = \frac{\max m(t) - \min m(t)}{2 + \max m(t) + \min m(t)}$

For sinusoidal modulation: $m(t) = m_a\cos(2\pi f_m t)$
$\mu = m_a = \frac{A_{max} - A_c}{A_c}$

Modulation Percentage:
$m\% = \mu \times 100\%$

Envelope Detection Condition

To avoid distortion in envelope detection:
$f_c >> \frac{1}{2\pi RC} >> f_m$

Where:
- f_c = Carrier frequency
- f_m = Maximum message frequency
- RC = Time constant of envelope detector

Power Calculations

Total Power:
$P_T = P_c + P_{SB} = \frac{A_c^2}{2} + \frac{A_c^2}{2}\overline{m^2(t)}$

Where:
- $P_c = \frac{A_c^2}{2}$ = Carrier power
- $P_{SB} = \frac{A_c^2}{2}\overline{m^2(t)}$ = Sideband power

Efficiency:
$\eta = \frac{P_{SB}}{P_T} = \frac{\overline{m^2(t)}}{1 + \overline{m^2(t)}}$

For sinusoidal modulation with $m(t) = m_a\cos(2\pi f_m t)$:
$\eta = \frac{m_a^2}{2 + m_a^2}$

AM Spectrum

$S_{AM}(f) = \frac{A_c}{2}[\delta(f - f_c) + \delta(f + f_c)] + \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]$

Bandwidth:
$B_{AM} = 2f_m$

Where f_m is the maximum frequency in $m(t)$.

Double Sideband Suppressed Carrier (DSB-SC)

Complex Envelope:
$g_{DSB-SC}(t) = A_c\, m(t)$

Time Domain:
$s_{DSB-SC}(t) = A_c\, m(t)\cos(2\pi f_c t)$

Power:
$P_{DSB-SC} = \frac{A_c^2}{2}\overline{m^2(t)}$

Efficiency: 100% (no carrier power)

Spectrum:
$S_{DSB-SC}(f) = \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]$

Single Sideband (SSB)

Transmits only one sideband (upper or lower) of the AM signal.

Upper Sideband (USB):
$s_{SSB-USB}(t) = A_c[m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t)]$

Lower Sideband (LSB):
$s_{SSB-LSB}(t) = A_c[m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t)]$

Where $\hat{m}(t)$ is the Hilbert transform of $m(t)$.

Bandwidth:
$B_{SSB} = f_m$

Power:
$P_{SSB} = \frac{A_c^2}{2}\overline{m^2(t)}$

Product Detection

Used for DSB-SC and SSB demodulation:

$s_d(t) = s(t) \cdot 2\cos(2\pi f_c t)$

After lowpass filtering:
$m_o(t) = A_c\, m(t)$

Lecture 18: Angle Modulation (FM & PM)

General Form

Angle modulation uses the **phase** of $g(t)$, not the amplitude. Information is carried in the phase/frequency.

Complex Envelope:
$$g(t) = A_c e^{j\theta(t)}$$

Bandpass Signal:
$$s(t) = A_c\cos(2\pi f_c t + \theta(t))$$

Phase Modulation (PM)

Phase directly proportional to message:
$$\theta(t) = k_p m(t)$$

Where k_p is the phase sensitivity constant (rad/V).

PM Signal:
$$s_{PM}(t) = A_c\cos(2\pi f_c t + k_p m(t))$$

Instantaneous Phase:
$$\phi_i(t) = 2\pi f_c t + k_p m(t)$$

Instantaneous Frequency:
$$f_i(t) = \frac{1}{2\pi}\frac{d\phi_i(t)}{dt} = f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$$

Instantaneous Frequency Deviation:
$$\Delta f(t) = f_i(t) - f_c = \frac{k_p}{2\pi}\frac{dm(t)}{dt}$$

Peak Frequency Deviation:
$$\Delta F = \frac{k_p}{2\pi}\max\left|\frac{dm(t)}{dt}\right|$$

Frequency Modulation (FM)

Frequency deviation proportional to message:
$$\Delta f(t) = k_f m(t)$$

Where k_f is the frequency sensitivity constant (Hz/V).

Instantaneous Frequency:
$$f_i(t) = f_c + k_f m(t)$$

Instantaneous Phase:
$$\theta(t) = 2\pi\int_0^t k_f m(\tau)\, d\tau = 2\pi k_f\int_0^t m(\tau)\, d\tau$$

FM Signal:
$$s_{FM}(t) = A_c\cos\left(2\pi f_c t + 2\pi k_f\int_0^t m(\tau)\, d\tau\right)$$

Peak Frequency Deviation:
$$\Delta F = k_f \max|m(t)|$$

Modulation Index

For PM:
$$\beta_{PM} = k_p \max|m(t)|$$

For FM:
$$\beta_{FM} = \frac{\Delta F}{f_m} = \frac{k_f \max|m(t)|}{f_m}$$

Where f_m is the maximum frequency in $m(t)$.

For sinusoidal modulation: $m(t) = A_m\cos(2\pi f_m t)$

PM:

$\beta_{PM} = k_p A_m$

FM:
$\beta_{FM} = \frac{k_f A_m}{f_m} = \frac{\Delta F}{f_m}$

Bessel Function Expansion (FM/PM Spectra)

For sinusoidal modulation, the FM/PM signal can be expanded using Bessel functions:

$s(t) = A_c\sum_{n=-\infty}^{\infty} J_n(\beta)\cos[2\pi(f_c + nf_m)t]$

Where $J_n(\beta)$ are Bessel functions of the first kind of order n.

Spectrum:
$S(f) = \frac{A_c}{2}\sum_{n=-\infty}^{\infty} J_n(\beta)[\delta(f - f_c - nf_m) + \delta(f + f_c + nf_m)]$

Power:
$P = \frac{A_c^2}{2}\sum_{n=-\infty}^{\infty} J_n^2(\beta) = \frac{A_c^2}{2}$

Carson's Rule (FM Bandwidth Approximation)

Approximate Bandwidth:
$B_{FM} \approx 2(\Delta F + f_m) = 2(\beta + 1)f_m$

For large β (wideband FM):
$B_{FM} \approx 2\Delta F$

For small β (narrowband FM):
$B_{FM} \approx 2f_m$

FM vs PM Comparison

Property	FM	PM		
Information in	Frequency deviation	Phase		
Instantaneous frequency	$f_c + k_f m(t)$	$f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$		
Instantaneous phase	$2\pi k_f\int m(t)dt$	$k_p m(t)$		
Bandwidth	Depends on ΔF and f_m	Depends on $\frac{dm(t)}{dt}$		
Noise immunity	Better for high frequencies	Better for low frequencies		
Modulation index	$\beta = \frac{\Delta F}{f_m}$	$\beta = k_p \max\	m(t)\	$

Key Relationships

Converting PM to FM:
If $m_{PM}(t) = \frac{dm_{FM}(t)}{dt}$, then PM with $m_{PM}(t)$ is equivalent to FM with $m_{FM}(t)$.

Converting FM to PM:
If $m_{FM}(t) = \int m_{PM}(t)dt$, then FM with $m_{FM}(t)$ is equivalent to PM with $m_{PM}(t)$.

Summary of Key Concepts

Lecture 13: Bandpass Signals

- Complex envelope $g(t) = x(t) + jy(t) = R(t)e^{j\theta(t)}$

• Simplifies high-frequency signal processing
• Power calculations using envelope
• Spectrum relationships

Lecture 17: Amplitude Modulation

- AM: *g(t) = A_c[1 + m(t)]*, information in amplitude
• Modulation index and percentage
• Power and efficiency calculations
• DSB-SC: Suppressed carrier, 100% efficiency
• SSB: Single sideband transmission, half bandwidth

Lecture 18: Angle Modulation

• FM/PM: Information in phase/frequency
• Modulation index β
• Bessel function expansion for spectra
• Carson's rule for bandwidth approximation
• FM vs PM characteristics and relationships

This summary covers the essential formulas and concepts from Lectures 13, 17, and 18 for Communication Systems.